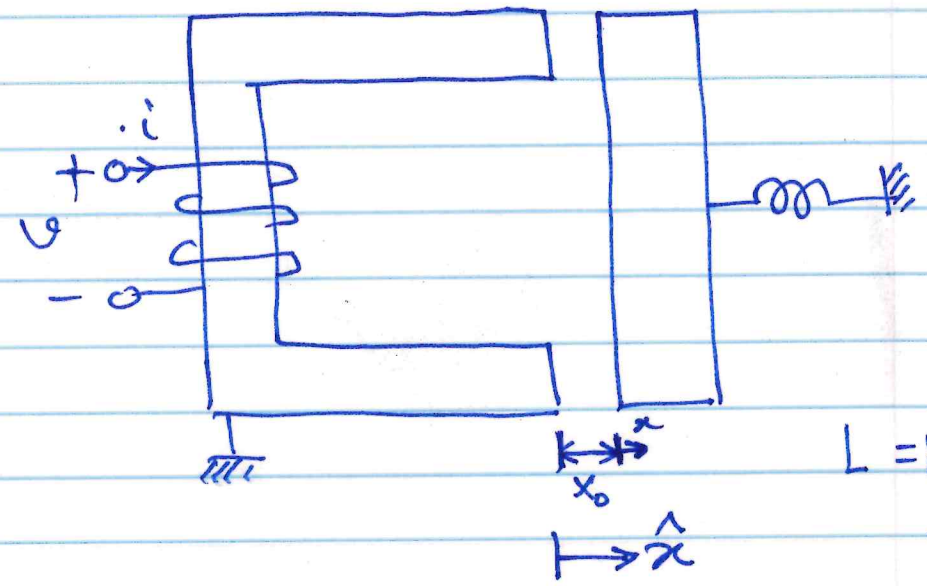
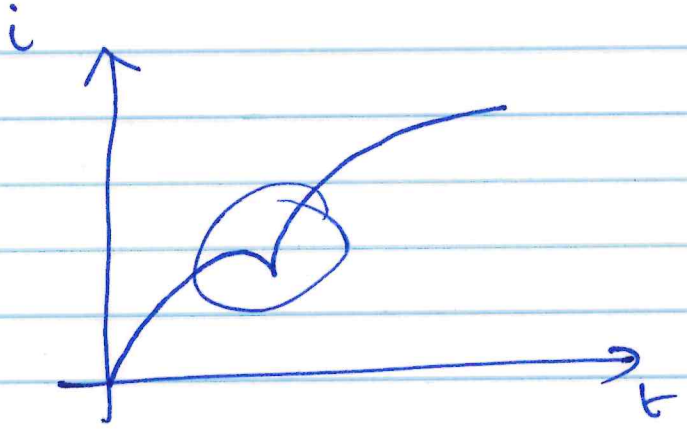
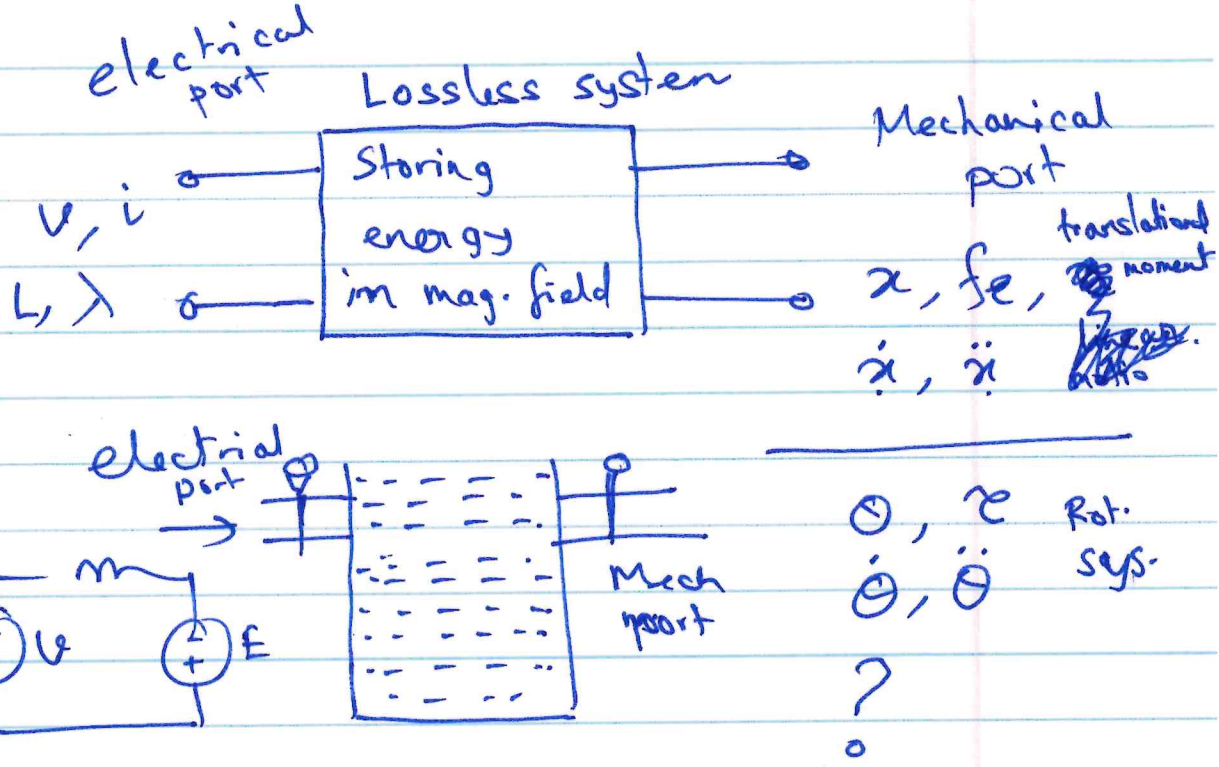


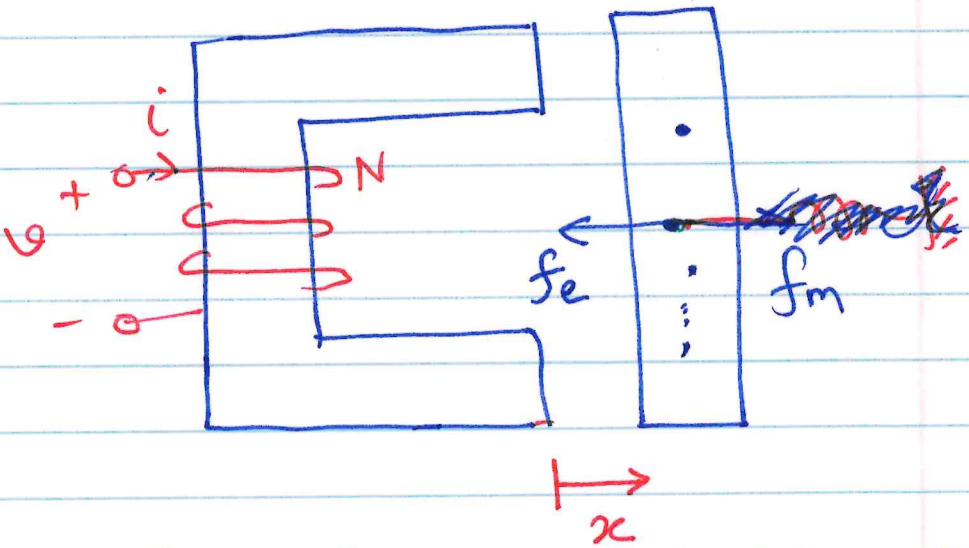
$$L = \frac{\mu_0 N^2 A}{2(X_0 + x)}$$

$$v = L(x)\frac{di}{dt} + i\frac{\partial L(x)}{\partial x}\frac{dx}{dt}$$

$$L(x) = \frac{N^2}{2R_0}$$







f_e = force of electrical origin.

$f_m dx$

$$p_e(t) = v(t) \cdot i(t)$$

$$= \frac{d\lambda}{dt} i(t)$$

$$E_e = \int p_e dt = \int i \, d\lambda$$

$$E_m = \int f_m dx$$

$$\omega_m = \int i \, d\lambda + \int f_m dx \qquad f_x \cdot x$$

$$f_e = -f_m$$

Physics

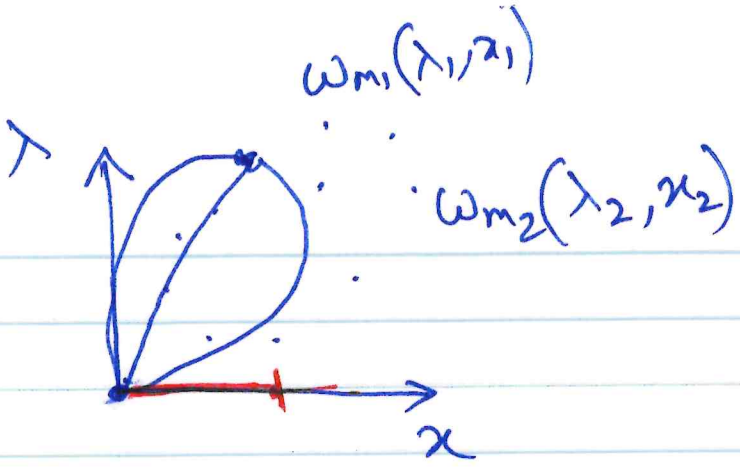
$$\boxed{d\omega_m = i \, d\lambda - f_e \, dx}$$

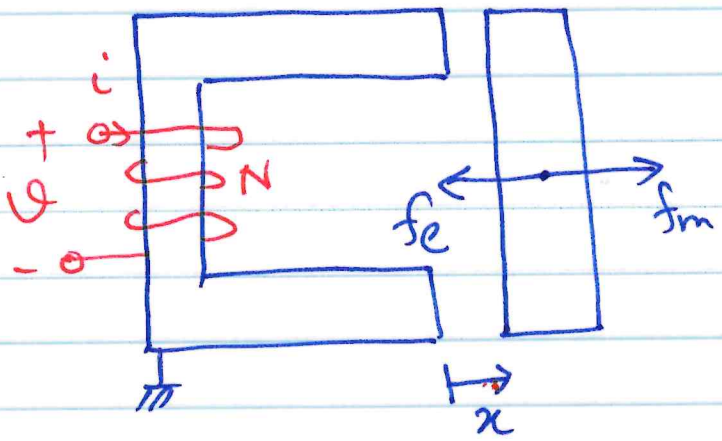
$$\boxed{\omega_m(\lambda, x)}$$

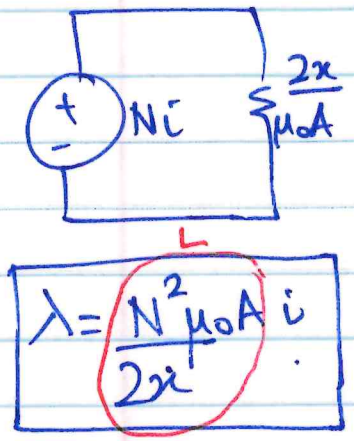
Math

$$\boxed{d\omega_m = \frac{\partial \omega_m}{\partial \lambda} \cdot d\lambda + \frac{\partial \omega_m}{\partial x} \cdot dx}$$

$$\left.\frac{\partial \omega_m}{\partial \lambda}\right|_{x=\text{const}} = i \qquad \left.\frac{\partial \omega}{\partial x}\right|_{\lambda=\text{const}} = -f_e$$







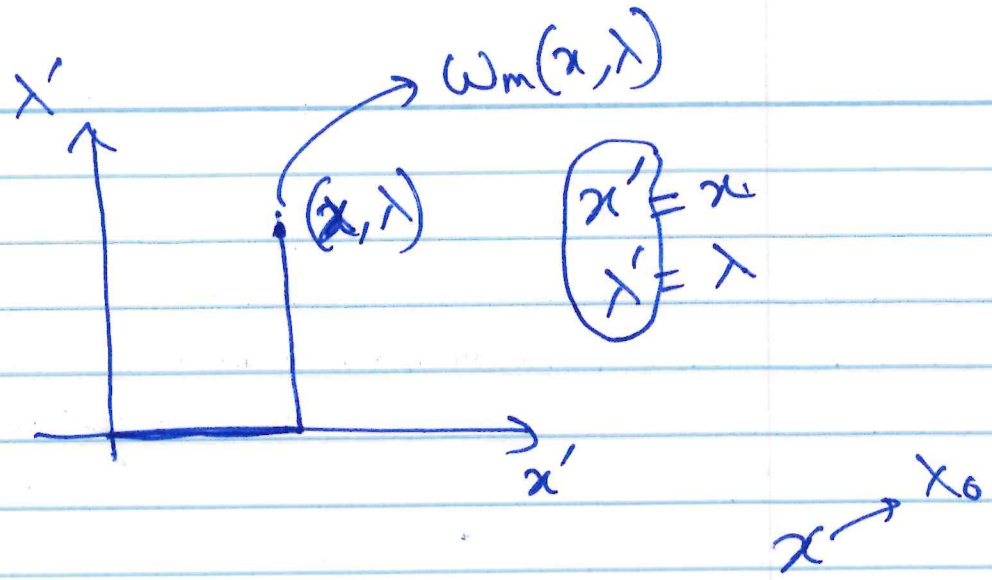
$$d\omega_m = i\,d\lambda - f_e\,dx$$

$$d\omega_m = \left.\frac{\partial \omega_m}{\partial \lambda}\right|_{x=\text{const}} \cdot d\lambda + \left.\frac{\partial \omega_m}{\partial x}\right|_{\lambda=\text{const}} \cdot dx$$

$$\left.\frac{\partial \omega_m}{\partial \lambda}\right|_{x=\text{const}} = i \qquad \left.\frac{\partial \omega_m}{\partial x}\right|_{\lambda=\text{const}} = -f_e$$

$$\omega_m^{(\lambda', x')} = \int d\omega_m = \int_0^{\lambda} i\,d\lambda$$

$$\omega_m(\lambda^*, x^*) = \int_0^{\lambda} \frac{\lambda'}{L(x)} \cdot d\lambda' = \frac{1}{L(x)} \frac{\lambda^2}{2}$$


x'
$\omega_m(x,\lambda)$
(x,λ)
$x' = x$
$\lambda' = \lambda$
x'
$x \to x_0$